MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE, SHIELDED, (FOR MIL-C-81511 SERIES 4 CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the contacts described herein shall consist of this document and the latest issue of Specification MIL-C-39029.

INACTIVE FOR NEW DESIGN

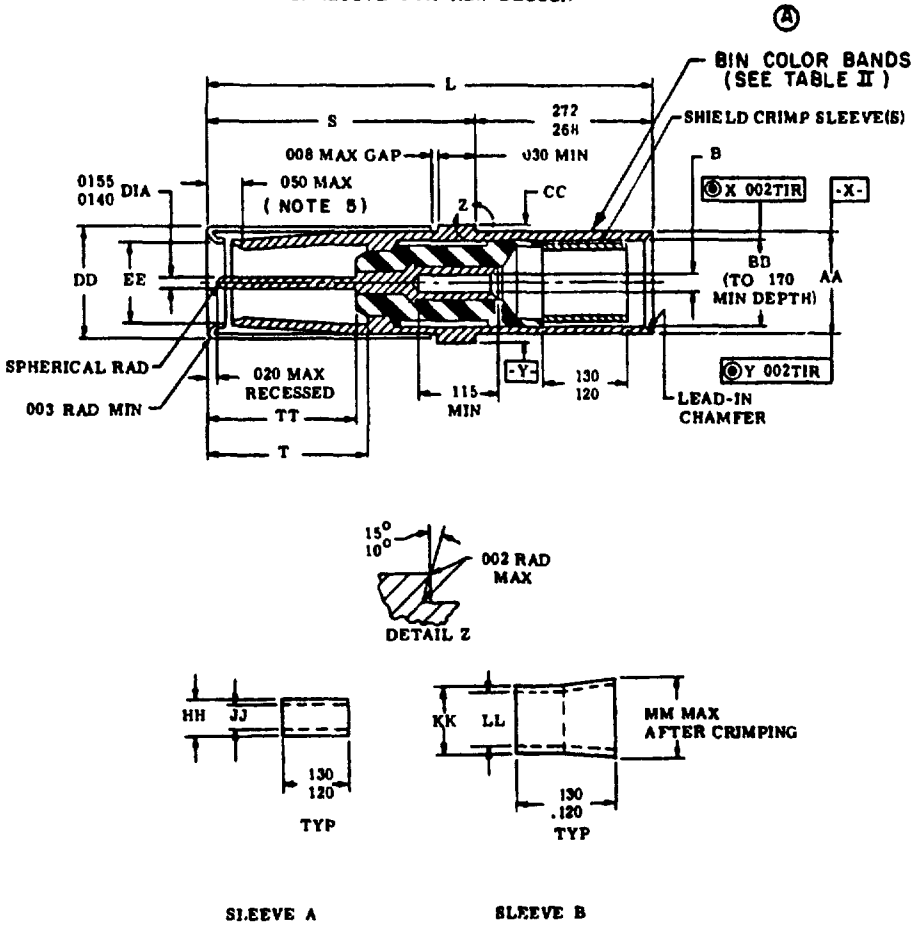


FIGURE 1. Connector contact.

FSC 5935

Source: https://assist.dla.mil -- Downloaded: 202610-21T18:32Z
Check the source to verify that this is the current version before use.

TABLE I. Dimensions.

BIN code	B MIN	S	L (REF)	T MIN	AA	BB MIN	CC	DD MAX	EE	TT MIN	SHIELD CRIMP SLEEVES					
												SLEEVE A		SLEEVE B		
											SLEEVES REQUIRED	HH MAX	JJ MIN	KK MAX	LL MIN	MM MAX
187	.034	.396	.664 (16.87)	.240	.094	.081	.108 (2.74)	.108	.0750	.215	A & B	.0535	.0455	.080	.072	.080
		.392			.091		.106 (2.69)		.0725		B	---	---	.080	.072	.080
188	.022	.386	.654 (16.61)	.225	.153	.129	.1695 (4.31)	.159	.1210	.200	NONE	---	---	---	---	---
		.382			.150		.1675 (4.25)		.1175		B	---	---	.128	.109	.128
189	.034	.386	.654 (16.61)	.225	.153	.129	.1695 (4.31)	.159	.1210	.200	B	---	---	.128	.113	.128
											A	.103	.081	---	---	---
		.382			.150		.1675 (4.25)		.1175		B	---	---	.128	.118	.128
											A & B	.099	.091	.128	.118	.128

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses for overall length and diameter only.
4. Dimensions shown apply after plating.
5. The distance between the end of the socket contact sleeve and a point at which a gage having a square face and the same basic diameter as the mating pin contact, first engages the contact spring member.
6. Axial concentricity: All diameters to be concentric within .004 TIR except as noted.

Source: https://assist.dla.mil -- Downloaded: 2026/10-21T18:32Z
Check the source to verify that this is the current version before use.

ASSEMBLY PROCEDURE (REF)

1. Cut cable to dimension shown in table II. Multiple shield cable must have outer jacket and outer shield removed before stripping.

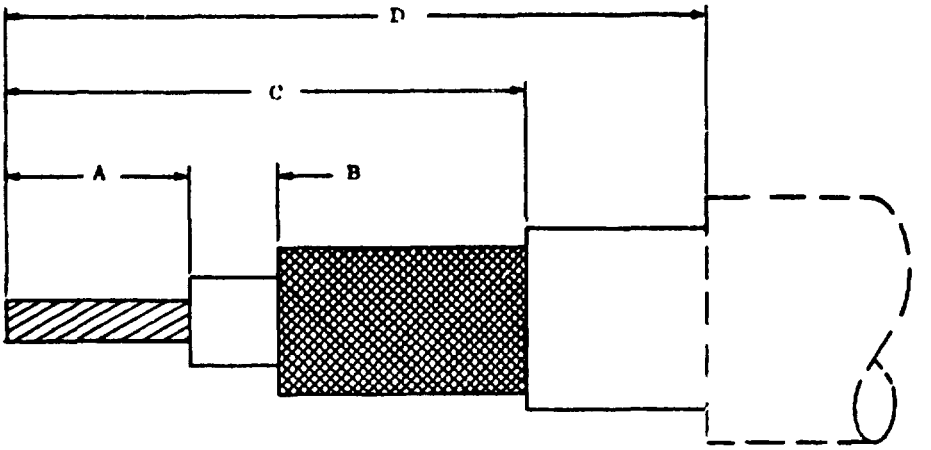


TABLE II. Cable accommodation dimension.

BIN CODE	CABLE ACCOMMODATION	A	B +.030 -.000	C +.030 -.000	D MIN
187	M81044/18-22 SHIELDED GRUMMAN GC18K1-22	.110 +.030 -.000	.065	.325	
	M27500-22RC1S6 REVERE WH95623 GRUMMAN GC875FW22-2 1/				
188	RG180B/U MICRODOT 293-3922 GRUMMAN GC875GA1	135 +.000 -.030	.040	.345	1.110
	RAYCHEM 9530D5117 GRUMMAN GC875GC1	.130 +.000 -.030	045	.345	1.110
189	M81044/18-22 SHIELDED GRUMMAN GC18K1-22	see item 1.a.			
	M27500-22RC1S6 REVERE WH95623 GRUMMAN GC875FW22-2	see item 1.b.			
	MICRODOT 250-4070 GRUMMAN GC875FY1	.135 +.000 -.030	.040	.345	1.110
	RAYCHEM 5022E5111 GRUMMAN GC875GB1 1/	.130 +.000 -.030	.045	.345	1.110

1/ See item 1.c.

Slide shield crimp sleeve over cable to cut end of outer jacket. M39029/21-02 when used with RG180B/U, MICRODOT 293-3922 or GRUMMAN GC875GA1 cable uses no shield crimp sleeve.

FIGURE 2. Assembly procedure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:32Z
Check the source to verify that this is the current version before use.

ASSEMBLY PROCEDURE (REF)

1.a. When using M81044/18-22 or GRUMMAN GC18K1-22 cable with M39029/21-03 contact, cut cable jacket to the shield as shown in figure A. Fold shield back tightly against cut end of jacket and trim insulation from center conductor as shown in figure B. Install shield crimp sleeve "B" as shown in figure C Trim shield ends flush with shield crimp sleeve.

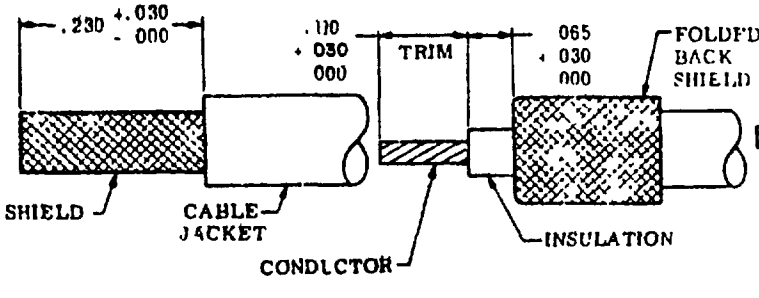


FIGURE A

FIGURE B

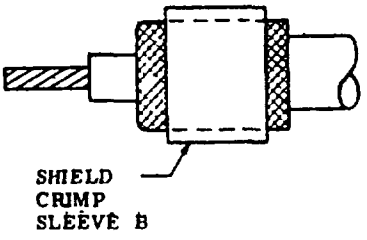


FIGURE C

1.b. When using M27500-22RC1S6, REVERE WH95623 or GRUMMAN GC875FW22-2 cable with M39029/21-03 contact, cut cable jacket to shield as shown on figure D Place shield crimp sleeve "A" over shield and butt against cut end of cable jacket as shown in figure E Fold shield back tightly over the shield crimp sleeve and trim shield ends flush with the shield crimp sleeve. Trim insulation from center conductor as shown in figure F

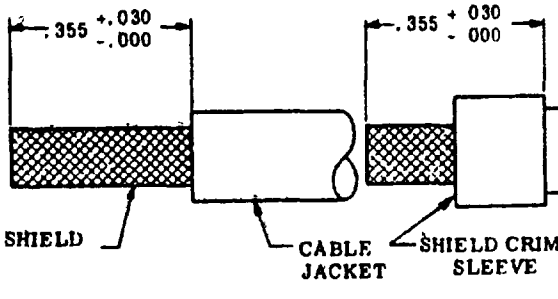


FIGURE D

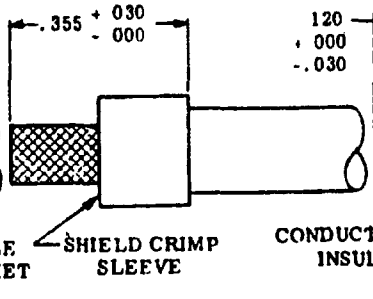


FIGURE E

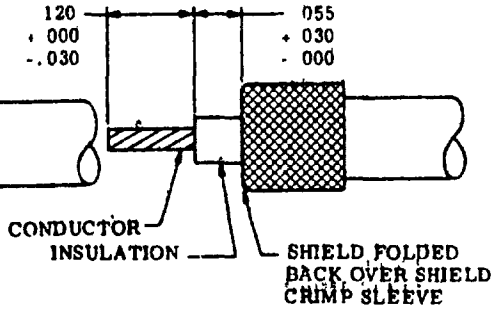


FIGURE F

FIGURE 2. Assembly procedure. -Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:32Z
Check the source to verify that this is the current version before use.

ASSEMBLY PROCEDURE (REF)

1.c. When using M81044/18-22 or GRUMMAN GC18K1-22 cable with M39029/21-01 contact or RAYCHEM 5022E5111 or GRUMMAN GC875GB1 cable shield crimp sleeve "A" and "B" are used. Insert the shield crimp sleeve "A" between the insulation and the shield until it is flush with the end of the shield as in figure G. Slide shield crimp sleeve "B" back over the shield until it is flush with both the shield and shield crimp sleeve "A".

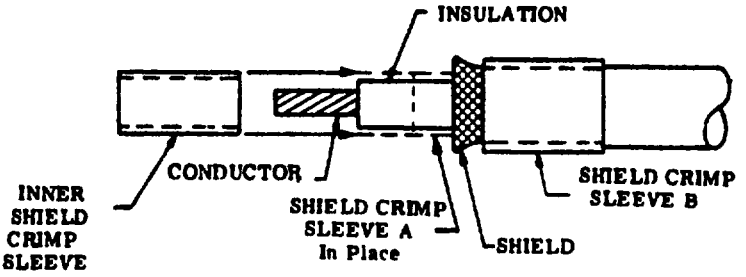
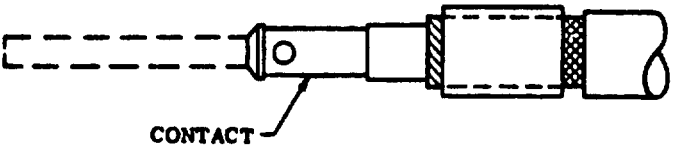


FIGURE G

2. Place inner contact on cable conductor.



2.a. M39029/21-01. Crimp inner contact in accordance with table IV. Push the contact insulator over the inner contact until the inner contact bottoms in the contact insulation.

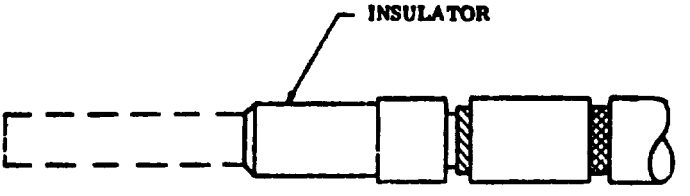
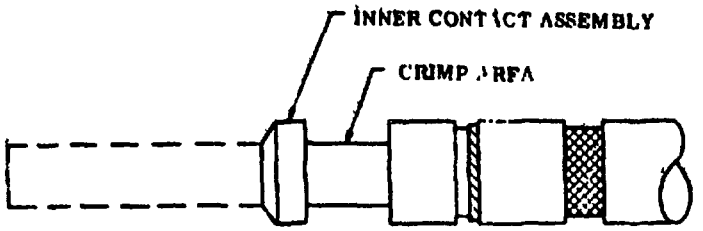


FIGURE 2. Assembly procedure. -Continued

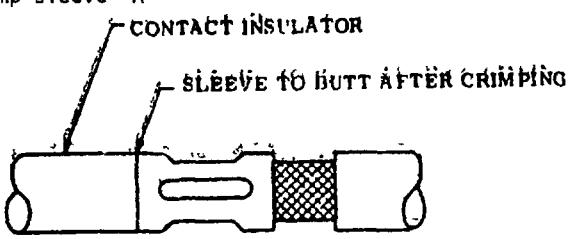
Source: https://assist.dla.mil -- Downloaded: 202610-21T18:32Z
Check the source to verify that this is the current version before use.

ASSEMBLY PROCEDURE (REF)

2.b. M39029/21-02 and M39029/21-03. Crimp inner contact through contact insulator using tooling in accordance with table IV.



3. Butt the shield crimp sleeve against the contact insulator and crimp the shield crimp sleeve in accordance with table IV. This crimp is not required when using M39029/19-02 contact with cables RG-180B/U, MICRODOT 293-3922, and GRUMMAN GC875GA1 which uses no shield crimp sleeves; or M39029/19-02 contact with cables M27500-22RC1S6, REVERE WH95623, and GRUMMAN GC875FW-22-2, which uses only shield crimp sleeve "A"



4. Install the inner contact and cable assembly into the outer contact. Push on the cable until the inner contact seat within the outer contact. Press cable into the outer contact while crimping the outer contact using tooling in accordance with table IV: Inspect for proper assembly in accordance with the figure below.

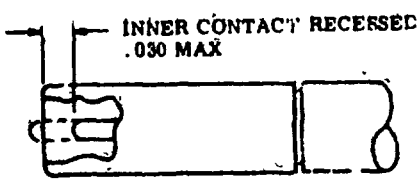


FIGURE 2. Assembly procedure. -Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-21T18:32Z
Check the source to verify that this is the current version before use.

TABLE III. Design characteristics.

BIN code	Color bands			Contact cavity size	Cables accommodated 1/	Sleeves required	Type	Class
	1st	2nd	3rd					
187	Brown	Gray	Violet	16	M81044/18-22 SHIELDED GRUMMAN GC18K1-22	A & B	I	E
					M27500-22RC1S6 REVERE MH95623 2/ GRUMMAN GC875FW22-2	B		
188	Brown	Gray	Gray	12	RG-180B/U 3/ MICRO DOT 293-3922 GRUMMAN GC875GA1	None		
					RAYCHEM 9530D5117 GRUMMAN GC875GC1	B		
189	Brown	Gray	White	12	M81044/18-22 SHIELDED GRUMMAN GC18K1-22	B		
					M27500-22RC1S6 REVERE MH95623 2/ GRUMMAN GC875FW-22-2	A		
					MICRO DOT 250-4070 GRUMMAN GC875FY2	B		
					RAYCHEM SD22E5111 GRUMMAN GC875GB1	A & B		

1/ or equivalent
2/ Revere MH95623 is a two conductor cable: One AWG22 wire and one AWG22 shielded conductor conforming to M27500-22RC1S6.
3/ RG-180B/U may be used when impedance matching is not required.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T18:32Z
Check the source to verify that this is the currentversion before use.

TABLE IV. Inner concact and shield crimp sleeve tooling.

BIN code	Inner contact tooling			Shield crimp sleeve tooling		
	Basic crimping tool	Positioner	Positioner setting	Basic crimping tool	Positioner	Positioner setting
187	M22520/2-01	M22520/2-28	4	M22520/20-01	M22520/20-02	3
	M22520/2-01	M22520/2-28	4	M22520/20-01	M22520/20-02	4
188	M22520/2-01	M22520/2-29	1	M22520/22-01	M22520/22-02	6
	M22520/2-01	M22520/2-29	1	M22520/22-01	M22520/22-02	3
189	M22520/2-01	M22520/2-29	4	M22520/22-01	M22520/22-02	4
	M22520/2-01	M22520/2-29	4	M22520/22-01	M22520/22-02	5
	M22520/2-01	M22520/2-29	4	M22520/22-01	M22520/22-02	6
	M22520/2-01	M22520/2-29	4	M22520/22-01	M22520/22-02	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:32Z
Check the source to verify that this is the current version before use.

TABLE V Outer contact tooling

BIN code	Outer contact tooling			Installing 1/ tool	Removal tool 1/	
	Basic crimping tool	Positioner	Positioner setting		Wired contact	Unwired contact
187	M22520/19-01	M22520/19-03	3	MS3160-16	MS3160-16	MS3159-16
	M22520/19-01	M22520/19-03	4	MS3160-16	MS3160-16	MS3159-16
188	M22520/21-01	M22520/21-03	5	MS3160-12	MS3160-12	MS3159-16
	M22520/21-01	M22520/21-03	3	MS3160-12	MS3160-12	MS3159-16
189	M22520/21-01	M22520/21-03	4	MS3160-12	MS3160-12	MS3159-16
	M22520/21-01	M22520/21-03	6	MS3160-12	MS3160-12	MS3159-16
	M22520/21-01	M22520/21-03	6	MS3160-12	MS3160-12	MS3159-16
	M22520/21-01	M22520/21-03	5	MS3160-12	MS3160-12	MS3159-16

1/ Installing and removal tools used with MIL-C-81511 series 4 connectors.

TABLE VI. Test limits.

BIN code	Tensile strength (lbs) min		Contact engagement and separation forces (ounces)				Contact resistance (millivolts)		Dielectric withstanding voltage (VAC RMS)		Insulation resistance (megohms)
			Outer socket contact								
			Separation		Engagement						
	Inner contact	Outer contact	Min	Test pin dia	Max	Test pin dia	Inner contact	Outer contact	Sea level	70,000 ft (mated only)	
187	2.7	15	2.0	.0690 +.0000 -.0001	20.0	.0710 +.0001 -.0000	3.0	4.0	1000	300	1000
188	4.5		3.0	.1140 +.0000 -.0001	25.0	.1165 +.0001 -.0000		2.0			
189	2.7										

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:32Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions, design characteristics, and configuration See figure 1 and tables I and III. Each contact is supplied with all the inner sleeves necessary to terminate any one of the cables which it accommodates.

Tools: See tables IV and V.

Manufacturer's recommended assembly instructions to be shipped with unit package.

Assembly procedure: See figure 2.

Mating contacts. MIL-C-39029/19.

Preparation of samples: Contacts shall be wired as required using wire in accordance with table III (cables accommodated column).

Contact resistance Mated contact resistance at a load current of 1 0 ampere shall meet the requirements of table VI.

Contact engagement and separation force. The contact engagement and separation forces shall meet the requirements of table VI. Oversize probe damage is not required for outer socket contact.

Insulation resistance: The insulation resistance when measured between the crimped outer contact and the center conductor shall meet the requirements of table VI.

Dielectric withstanding voltage. The dielectric withstanding voltage with the contacts withstanding voltage with the contacts mated and unmated shall meet the requirements of table VI.

Tensile strength: The tensile strength shall meet the requirements of table VI.

QPL evaluating activity: Naval Weapons Support Center, Code 3074, Crane, Indiana 47522

Military part numbers: See table VII.

TABLE VII. Military part number.

BIN code	Military part number
187	M39029/21-01
188	M39029/21-02
189	M39029/21-03

Custodians:
Navy - AS

Review activities:
Navy - EC, SH, MC
DSA - ES

User activities:
Navy - OS

Preparing activity
Navy - AS

Agent:
DSA - ES

(Project 5935-N041-8)

Source: https://assist.dla.mil -- Downloaded: 20260-21T18:32Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:32Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Engineering Specifications and Standards Department(Code 93)
Naval Air Engineering Center
Lakehurst, NJ 08733

FOLD

Source: https://assist.dla.mil -- Downloaded: 202640-21T18:32Z
Check the source to verify that this is the currentversion before use.